

Coding & Solution

Date	04 July 2026
Team ID	XXXXXX
Project Name	Opticrop-Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

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Field	Details
Repository Link / URL	https://github.com/indravathiyalla8-coder
Programming Language(s)	Python, JavaScript, HTML, CSS, SQL
Framework(s) Used	Flask, Bootstrap, Pandas, Scikit-learn
Key Features Implemented	• Crop recommendation based on soil nutrients (N, P, K), temperature, humidity, pH and rainfall. • Machine learning model for accurate crop prediction. • User-friendly web interface for easy input and results. • Data validation and error handling. • Stores user history and recommendations.
Bundling (If any updates or patches included)	Includes all source code, trained model, dataset, and README file with setup instructions.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

- The code is well-structured, efficient, and follows best practices.
- All modules are properly tested and integrated.
- The project is maintained in a GitHub repository.
- GitHub Link: <https://github.com/indravathiyalla8-coder>